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Assignment 5: Predict the price of the Uber ride from a given pickup point to the agreed drop-off location. Perform following tasks: 1. Pre-process the dataset. 2. Identify outliers. 3. Check the correlation. 4. Implement linear regression and ridge, Lasso regression models. 5. Evaluate the models and compare their respective scores like R2, RMSE, etc. Dataset link: https://www.kaggle.com/datasets/yasserh/uber-fares-dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import StandardScaler
```

Load Dataset

```
df = pd.read_csv("uber.csv")
df.head()
```

	Unnamed: 0	key	fare_amount	pickup_datetime	pickup_longitude	pickup_latitude	dropoff_longitude	dropoff_
0	24238194	2015-05-07 19:52:06.0000003	7.5	2015-05-07 19:52:06 UTC	-73.999817	40.738354	-73.999512	4
1	27835199	2009-07-17 20:04:56.0000002	7.7	2009-07-17 20:04:56 UTC	-73.994355	40.728225	-73.994710	4
2	44984355	2009-08-24 21:45:00.00000061	12.9	2009-08-24 21:45:00 UTC	-74.005043	40.740770	-73.962565	4
3	25894730	2009-06-26 08:22:21.0000001	5.3	2009-06-26 08:22:21 UTC	-73.976124	40.790844	-73.965316	4
4	17610152	2014-08-28 17:47:00.000000188	16.0	2014-08-28 17:47:00 UTC	-73.925023	40.744085	-73.973082	4

Basic info

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 35725 entries, 0 to 35724
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Unnamed: 0         35725 non-null  int64
 1   key                35725 non-null  object
 2   fare_amount        35725 non-null  float64
 3   pickup_datetime    35725 non-null  object
 4   pickup_longitude   35724 non-null  float64
 5   pickup_latitude    35724 non-null  float64
 6   dropoff_longitude  35724 non-null  float64
 7   dropoff_latitude   35724 non-null  float64
 8   passenger_count    35724 non-null  float64
dtypes: float64(6), int64(1), object(2)
memory usage: 2.5+ MB
```

	Unnamed: 0	fare_amount	pickup_longitude	pickup_latitude	dropoff_longitude	dropoff_latitude	passenger_count
count	3.572500e+04	35725.000000	35724.000000	35724.000000	35724.000000	35724.000000	35724.000000
mean	2.765762e+07	11.394681	-72.561212	39.942645	-72.572503	39.943957	1.677024
std	1.604217e+07	10.085005	10.911110	6.047059	10.852786	6.043435	1.294493
min	4.800000e+01	0.000000	-748.016667	-74.015515	-737.916665	-74.008745	0.000000
25%	1.367902e+07	6.000000	-73.992033	40.734753	-73.991475	40.733846	1.000000
50%	2.757853e+07	8.500000	-73.981821	40.752563	-73.980170	40.752855	1.000000
75%	4.151460e+07	12.900000	-73.967196	40.767152	-73.963545	40.768076	2.000000
max	5.542169e+07	350.000000	40.774042	45.031653	40.828377	45.031598	6.000000

Data Preprocessing

```
df.drop(columns=['key'], inplace=True)
```

```
df.isnull().sum()
```

	0
Unnamed: 0	0
fare_amount	0
pickup_datetime	0
pickup_longitude	1
pickup_latitude	1
dropoff_longitude	1
dropoff_latitude	1
passenger_count	1

dtype: int64

```
df.dropna(inplace=True)
```

Convert Pickup dateTime

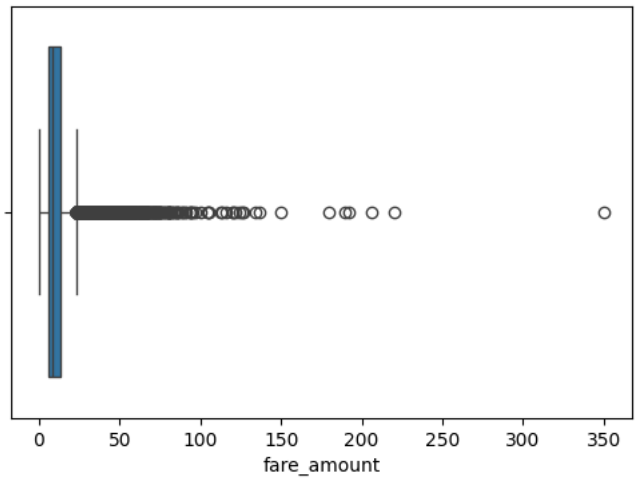
```
df['pickup_datetime'] = pd.to_datetime(df['pickup_datetime'])
df['hour'] = df['pickup_datetime'].dt.hour
df['day'] = df['pickup_datetime'].dt.day
df['month'] = df['pickup_datetime'].dt.month
df['weekday'] = df['pickup_datetime'].dt.weekday
df.drop(columns=['pickup_datetime'], inplace=True)
```

Remove Invalid Coordinates

```
df = df[
    (df['fare_amount'] > 0) &
    (df['pickup_latitude'].between(-90, 90)) &
    (df['pickup_longitude'].between(-180, 180)) &
    (df['dropoff_latitude'].between(-90, 90)) &
    (df['dropoff_longitude'].between(-180, 180))]
```

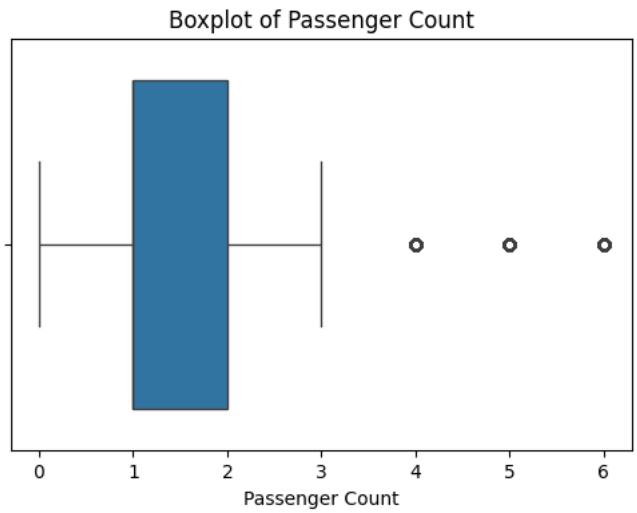
Outlier Detection

```
plt.figure(figsize=(6,4))
sns.boxplot(x=df['fare_amount'])
plt.show()
```




```
plt.figure(figsize=(6,4))
sns.boxplot(x=df['passenger_count'])
plt.title("Boxplot of Passenger Count")
```

```
plt.xlabel("Passenger Count")
plt.show()
```



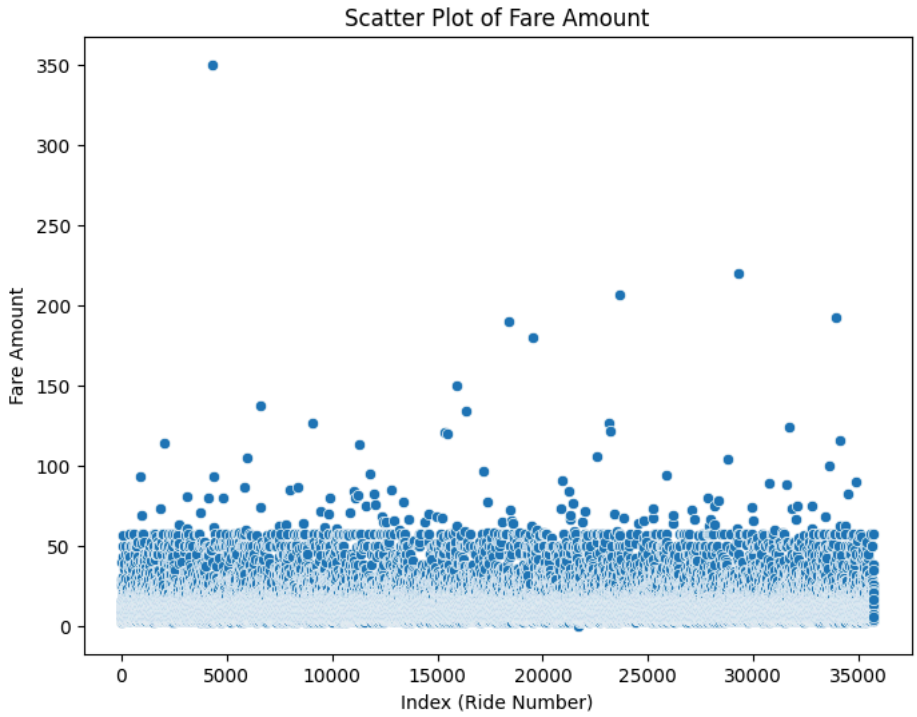
Check Correlation

```
df.corr()
```

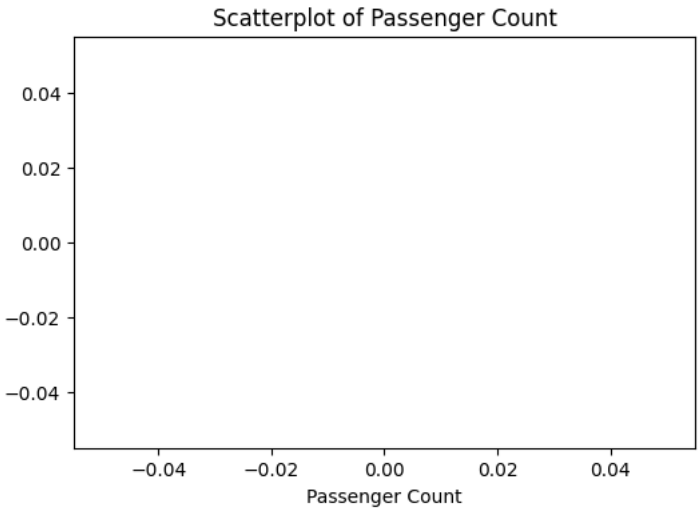
	Unnamed: 0	fare_amount	pickup_longitude	pickup_latitude	dropoff_longitude	dropoff_latitude	passenger_c
Unnamed: 0	1.000000	0.004087	0.001976	-0.001319	0.002842	-0.001030	0.00
fare_amount	0.004087	1.000000	0.029128	-0.028300	0.028971	-0.028348	0.01
pickup_longitude	0.001976	0.029128	1.000000	-0.978463	0.950938	-0.937102	0.00
pickup_latitude	-0.001319	-0.028300	-0.978463	1.000000	-0.939256	0.959943	-0.00
dropoff_longitude	0.002842	0.028971	0.950938	-0.939256	1.000000	-0.980667	0.00
dropoff_latitude	-0.001030	-0.028348	-0.937102	0.959943	-0.980667	1.000000	-0.00
passenger_count	0.008588	0.012631	0.003077	-0.005112	0.003888	-0.006214	1.00
hour	0.001319	-0.005083	-0.003885	0.005694	-0.003115	0.004507	0.01
day	0.001429	0.005072	0.008175	-0.011882	0.006532	-0.010220	0.00
month	0.006286	0.026608	-0.013404	0.013335	-0.011702	0.012254	0.00
weekday	-0.006678	0.006835	-0.009954	0.007410	-0.010106	0.006766	0.02

Scatter Plot

```
plt.figure(figsize=(8,6))
sns.scatterplot(
    x=df.index,
    y=df['fare_amount']
)
plt.title("Scatter Plot of Fare Amount")
plt.xlabel("Index (Ride Number)")
plt.ylabel("Fare Amount")
plt.show()
```




```
plt.figure(figsize=(6,4))
sns.scatterplot(x=df['passenger_count'])
plt.title("Scatterplot of Passenger Count")
plt.xlabel("Passenger Count")
plt.show()
```

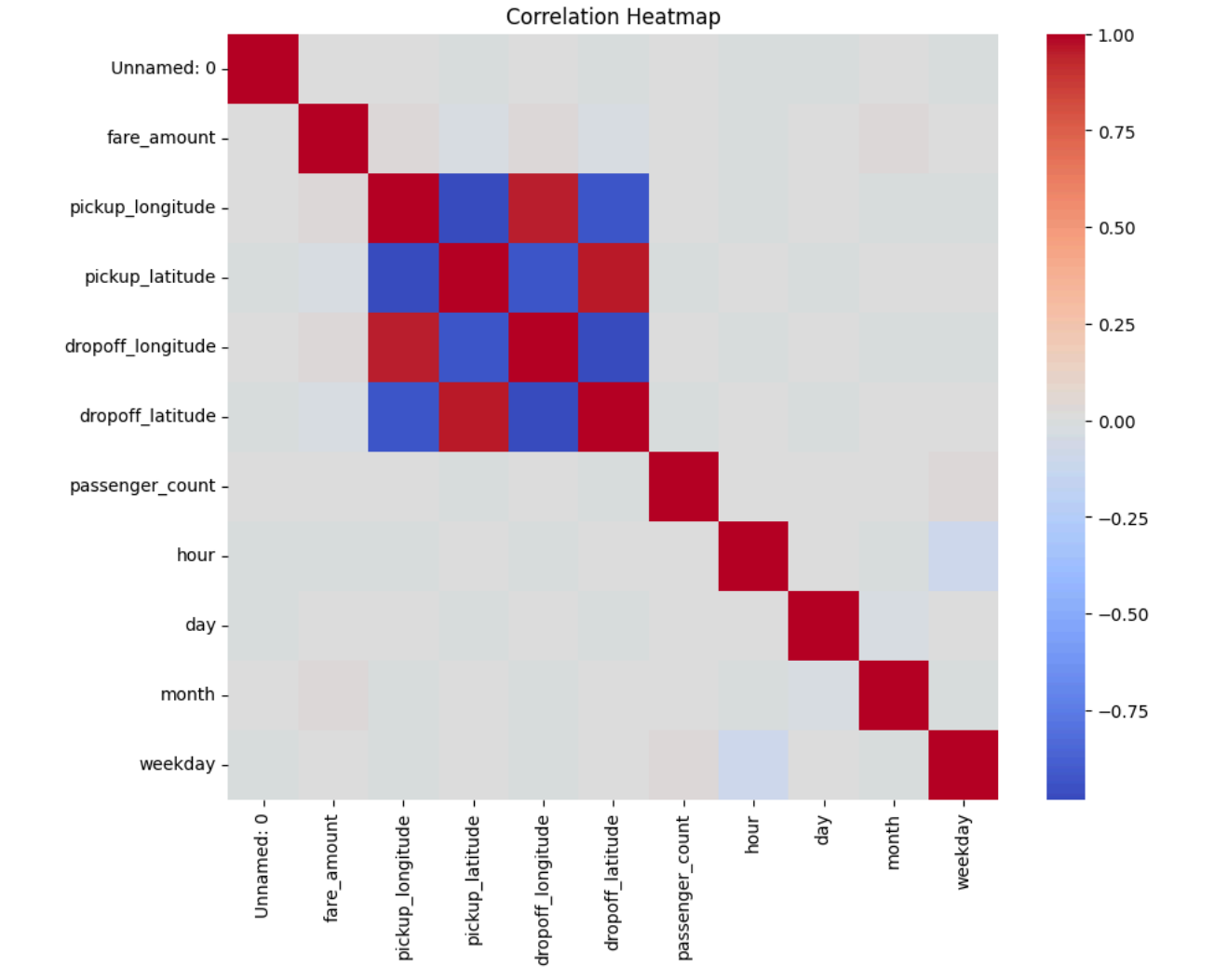



```
Q1 = df['fare_amount'].quantile(0.25)
Q2 = df['fare_amount'].quantile(0.50)
Q3 = df['fare_amount'].quantile(0.75)
IQR = Q3 - Q1
Q1, Q2, Q3, IQR
```

```
(np.float64(6.0), np.float64(8.5), np.float64(12.9), np.float64(6.9))
```

Correlation Analysis

```
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(), cmap='coolwarm', annot=False)
plt.title("Correlation Heatmap")
plt.show()
```



Linear Regression

```
import numpy as np
import matplotlib.pyplot as plt
x = np.array([2, 4, 6])
y_observed = np.array([50, 55, 65])
```

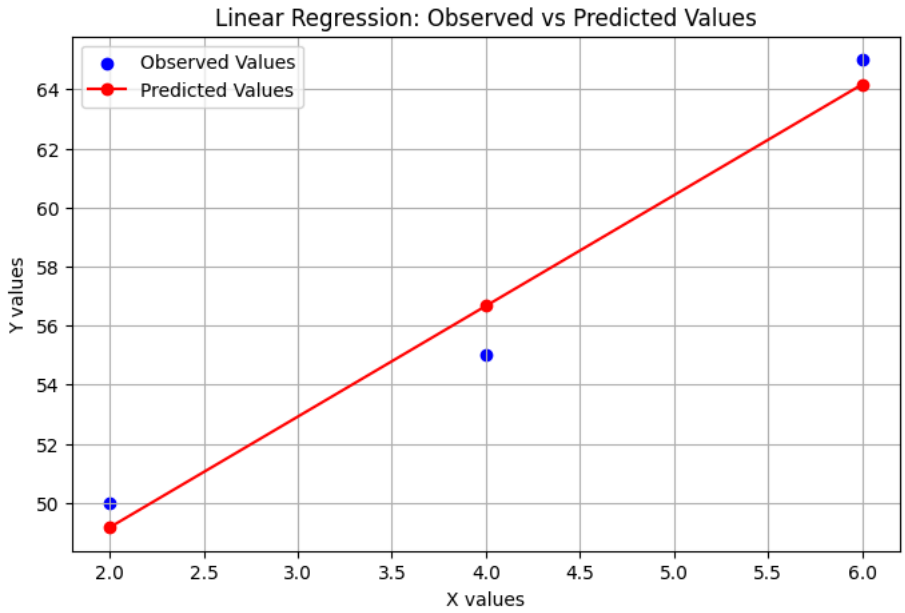
```
x_mean = np.mean(x)
y_mean = np.mean(y_observed)
```

```
b1 = np.sum((x - x_mean) * (y_observed - y_mean)) / np.sum((x - x_mean) ** 2)
b0 = y_mean - b1 * x_mean
```

```
y_predicted = b0 + b1 * x
print("b1 (Slope):", b1)
print("b0 (Intercept):", b0)
print("Observed values:", y_observed)
print("Predicted values:", y_predicted)
```

```
b1 (Slope): 3.75
b0 (Intercept): 41.666666666666664
Observed values: [50 55 65]
Predicted values: [49.16666667 56.66666667 64.16666667]
```

```
plt.figure(figsize=(8,5))
plt.scatter(x, y_observed, color='blue', label='Observed Values')
plt.plot(x, y_predicted, color='red', marker='o', label='Predicted Values')
plt.xlabel("X values")
plt.ylabel("Y values")
plt.title("Linear Regression: Observed vs Predicted Values")
plt.legend()
plt.grid(True)
plt.show()
```



Rainfall dataset

```
import numpy as np
import matplotlib.pyplot as plt
x = np.array([12.3, 13.7, 14.5, 11.2, 13.2, 14.1, 12.0])
y = np.array([6.25, 8.02, 8.42, 5.27, 7.21, 8.71, 5.68])
x_mean = np.mean(x)
y_mean = np.mean(y)
numerator = np.sum((x - x_mean) * (y - y_mean))
denominator = np.sqrt(np.sum((x - x_mean)**2) * np.sum((y - y_mean)**2))
r = numerator / denominator
print("Linear Correlation Coefficient (r):", r)
```

```
Linear Correlation Coefficient (r): 0.9807537506120445
```

```
# Slope
b1 = np.sum((x - x_mean) * (y - y_mean)) / np.sum((x - x_mean)**2)
# Intercept
b0 = y_mean - b1 * x_mean
print("Slope (b1):", b1)
print("Intercept (b0):", b0)
y_predicted = b0 + b1 * x
```

```
Slope (b1): 1.1153669724770643
Intercept (b0): -7.419770642201837
```

```
plt.figure(figsize=(8,5))
plt.scatter(x, y, label="Observed Values")
plt.scatter(x, y_predicted, label="Predicted Values")
plt.xlabel("Rainfall (cm)")
plt.ylabel("Yield (tonnes per acre)")
plt.title("Observed vs Predicted Values")
plt.legend()
plt.grid(True)
plt.show()
```